🔥 Fire and Smoke Detection using YOLOv8

📌 Project Overview

This project presents a **Fire and Smoke Detection System** using **YOLOv8 (You Only Look Once Version 8)**. The objective is to automatically detect fire and smoke from images using deep learning and computer vision techniques.

The project was developed in **Google Colab** using the **Ultralytics YOLOv8 framework** and trained on a custom Fire & Smoke dataset.

🎯 Objectives

- Detect **Fire** and **Smoke** in images.
- Train a custom YOLOv8 object detection model.
- Evaluate model performance on a validation dataset.
- Perform predictions on unseen test images.
- Demonstrate the application of deep learning in fire safety.

🛠️ Technologies Used

- Python
- Google Colab
- YOLOv8 (Ultralytics)
- PyTorch
- OpenCV
- NumPy
- Matplotlib
- KaggleHub

📂 Dataset

The dataset contains two object classes:

- 🔥 Fire
- 💨 Smoke

Dataset Structure:

```
data/
│
├── train/
│   ├── images/
│   └── labels/
│
├── val/
│   ├── images/
│   └── labels/
│
└── test/
    ├── images/
    └── labels/
```

🧠 Model

- Model: **YOLOv8 Nano (yolov8n)**
- Image Size: **640 × 640**
- Epochs: **1**
- Batch Size: **64**
- Optimizer: Default YOLOv8 Optimizer

🚀 Project Workflow

1. Install required libraries.
2. Download the dataset.
3. Configure `data.yaml`.
4. Load YOLOv8 model.
5. Train the model.
6. Validate model performance.
7. Predict fire and smoke on test images.
8. Save trained weights (`best.pt`).

📊 Output Files

After training, the following files are generated:

- `best.pt`
- `last.pt`
- `results.png`
- `confusion_matrix.png`

- `PR_curve.png`
- `P_curve.png`
- `R_curve.png`

📈 Applications

- Forest fire monitoring
- Industrial fire detection
- Smart surveillance systems
- CCTV-based safety monitoring
- Early warning systems

🔮 Future Improvements

- Increase the number of training epochs.
- Use a larger and more diverse dataset.
- Perform real-time webcam detection.
- Deploy the model using Flask or Streamlit.
- Optimize for edge devices such as Raspberry Pi.

📦 Requirements

```
ultralytics
torch
torchvision
opencv-python
numpy
matplotlib
PyYAML
kagglehub
```

Install all dependencies:

```
pip install -r requirements.txt
```

▶️ How to Run

1. Clone the repository.

2. Install the required packages.
3. Open the notebook in Google Colab.
4. Train the model.
5. Run prediction on test images.

📸 Sample Results

The trained model detects:

- 🔥 Fire
- 💨 Smoke

using bounding boxes with confidence scores.

👩‍💻 Author

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Deep Learning Project – Fire & Smoke Detection using YOLOv8

⭐ Acknowledgements

- Ultralytics YOLOv8
- PyTorch
- Google Colab
- KaggleHub